

PROJECT RC ACCELEROMETER

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Group 11

Introduction

An accelerometer is a device that measures the acceleration of motion, or vibration, of a dynamic structure. Remote controlled cars, often implement an accelerometer to detect impacts and collisions remotely, in case the vehicle would be damaged when beyond the view of the driver. By sensing the amount of dynamic acceleration, the device provides the driver with the necessary information, about the state of his car.

For this application, the accelerometer requires some specific characteristics. Firstly, the accelerometer must be mobile and provide a digital output that can be relayed to the driver in real time. In order to do so, the accelerometer will be connected to an Arduino UNO and ESP8266 Wi-Fi chip. Furthermore, a bandwidth is less important for this use, as the main objective of the accelerometer is to detect collisions, therefore a tolerance of $\pm 10\%$ is sufficient. Secondly, the driver must know the severity of the collision and with what acceleration his vehicle has collided. Consequently, the accelerometer must be sensitive enough to handle $\pm 2g$ force swings, as the car may experience sudden starts and stops [1]. Lastly, the device must be compact and lightweight so that the performance of the car remains unaffected when attached to the vehicle. To keep the design compact, the accelerometer should not be much larger or heavier than the Arduino board, as most RC cars will significantly suffer in performance beyond that point. To be exact, the accelerometer should be mounted on top of the Arduino board and not exceed its parameters, 75x60x30mm and 100 grams.

Design

For the above-mentioned application, a one-axis capacitive accelerometer is chosen, as it provides enough information to determine if a collision has occurred, and an electronic readout. It consists of a proof mass m that is attached to a fixed frame using a spring like object with spring constant k . On either side of the frame, the two capacitors are located, consisting of aluminium plates as they are highly conductive [3]. Consider the differential equation 1 modelling the dynamic behaviour of the system, where Z is the displacement relative to the frame, variable D is the damping coefficient and a is the acceleration of the mass.

$$-a = \frac{d^2 Z}{dt^2} + \frac{D}{m} \cdot \frac{dZ}{dt} + \frac{k}{m} \cdot Z \quad (1)$$

The change in voltage drop of the capacitor will be used to determine the displacement Z . So that the driver

can receive an electronic readout of the displacement Z with time. As the mass oscillates with regard to the movement of the car, it displaces the capacitors plates, causing a change in the voltage. The relation of a differential capacitor is given in equation 2 where V_r is the initial voltage.

$$Z = \frac{d\Delta V}{V_r} \quad (2)$$

To keep the voltage drop, relatively large, the initial displacement will be set to 1 mm. Instead of determining the mass m for a given spring constant, the mass will consist of two 37g brass blocks. The design will implement four cantilever beams instead of traditional beams, where the length and area of the beams will be altered to achieve the desired spring constant. This is described in the following equation 3

$$k = 4 \cdot \frac{3EI}{L^3} \quad (3)$$

Constant E is the material's young's modulus, I moment of area and L is the length. Whereas the shell of the accelerometer will be 3D printed, using thermoplastic polyurethane (TPU) as it is flexible enough to provide a measurable displacement and can be used in a 3D printer. TPU has a E modulus of 2410 N/mm² [2]. The length of the beams will be constricted to 2.5cm due to the size of the Arduino board.

Consider Fig.1 as an initial render of the accelerometer.

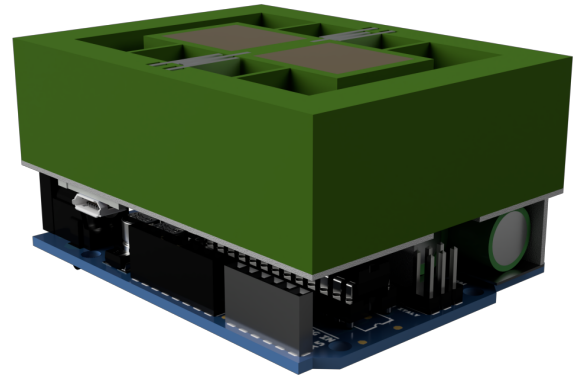


Figure 1: 3D render of initial accelerometer design. The frame is located inside and holds two brass blocks. It is connected to the outside shell using the cantilever beams. The two capacitors, have their plates in-between the beams.

Analysis

To judge the quality of the design, a time response was simulated from the results of a comparable experiment, where a skateboard with an accelerometer, crashed into

a wall after rolling down a hill. The Fourier transform was used on acceleration readings to determine what profound frequency (natural frequency) yielded the largest acceleration. With that natural frequency, a time response was plotted as seen in Fig.2.

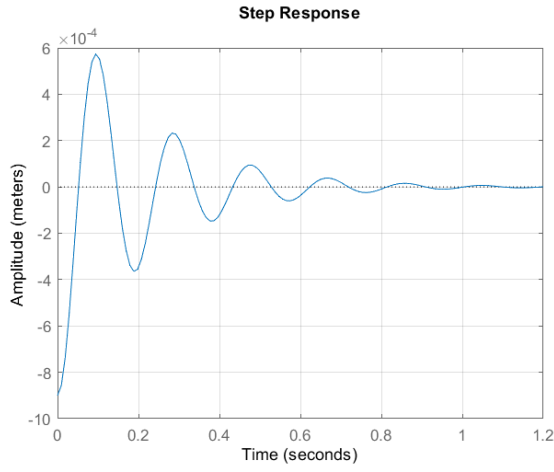


Figure 2: Step response simulated model with arbitrary damping

From the step response, the desired damping coefficient D was determined to be 4.2 Ns/m . Consider the table 1, containing the system parameters that the design should achieve. Using equation 1, a frequency response of the system can be modelled, as seen in Fig 3.

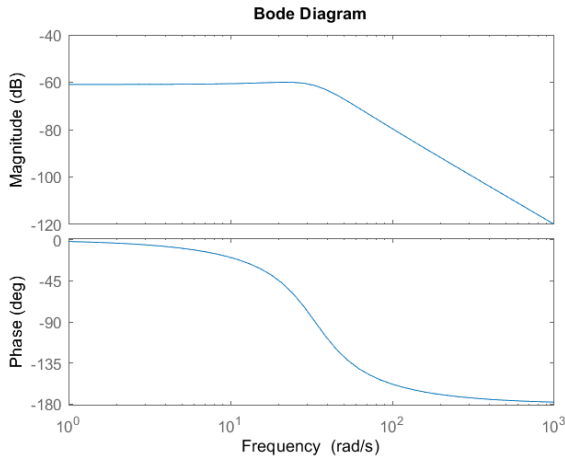


Figure 3: Bode plot of the simulated model with critical damping

The natural frequency and mass m were implemented to determine the theoretical spring constant needed for the design, 82.01 N/m . Using equation 3 the moment of area I was calculated to be 0.000443 cm^2 indicating that the design is feasible.

Realization

With all the necessary parameters, shown in Table.1, the design can be finalized. Fig. 4 is a schematic of the final accelerometer with the dimensions.

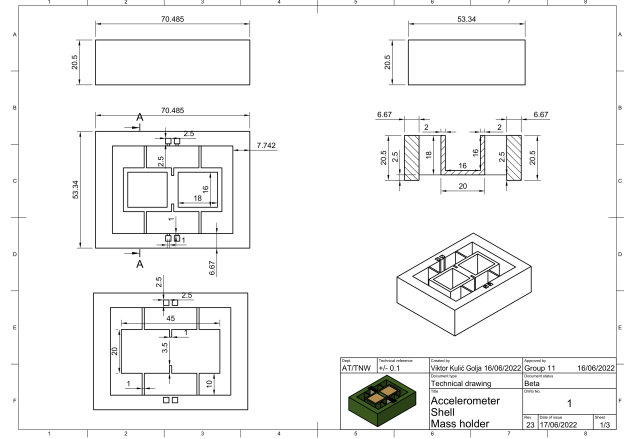


Figure 4: Schematic of accelerometer initial design with dimensions

The next steps will consist of sourcing the required material. This includes Arduino UNO, basic electric components used for circuitry which are provided by the study program, ESP8266 Wi-Fi chip and as well as the 3D print filament TPU. Afterwards, the design will be transferred to 3D printing software and printed. Lastly, when all components are available and assembled, the experimenting stage will begin to test the device's performance and make any modifications if need be. Some of the risks include; malfunctioning/damaged components, unstable construction and other unexpected risks. All of these factors contribute to time delays, considering those it is best to start experimenting, analysing the results and taking measures to improve the performance from early on.

Table 1: Design parameters and constraints

Variable	Value	Unit
M	74	g
D	4.2	Ns/m
k	82.01	N/m
I	0.000443	cm ²

References

- [1] Freescale Halbleiter Deutschland GmbH. Accelerometer Terminology Guide, May 2007.
- [2] Hyojeong Lee, Ran-I Eom, and Yejin Lee. Evaluation of the mechanical properties of porous thermoplastic polyurethane obtained by 3D printing for protective gear. *Advances in Materials Science and Engineering*, 2019:5838361, December 2019.
- [3] Soumendu Sinha, Snigdha Shakya, R. Mukhiya, Ram Gopal, and Bikram Pant. Design and simulation of mems differential capacitive accelerometer. 07 2014.